

CLAUDE_v1.5_FINAL_FREEZE.md

MonitorAI

Claude Development Constitution

Version 1.5 FINAL FREEZE

Status: Approved For Development

Project Identity

MonitorAI is an orthodontic image acquisition platform.

MonitorAI is NOT:

- A diagnostic AI
- A camera application
- A telemedicine platform

MonitorAI IS:

A standardized orthodontic image acquisition platform and future AI dataset generation platform.

Primary Objective

Collect highly standardized orthodontic images and metadata.

Every implementation decision must improve:

- Standardization
 - Repeatability
 - Clinical usefulness
 - Metadata quality
 - Future AI value
-

Technical Stack

Language

Swift 6

Framework

SwiftUI

Architecture

MVVM

Repository Pattern

Dependency Injection

Protocol-Oriented Design

Backend

FileMaker Server 19

FileMaker Data API

External Secure Storage

Mandatory Reading Order

Before coding any feature:

01_MasterSpec.md

↓

02_Workflow_v1.5.md

↓

03_UI_UX_Spec.md



04_FileMakerAPI.md



05_Architecture.md



06_DevelopmentWorkflow.md



Remaining Documents

Development Rules

Always implement one phase at a time.

Do not start future phases early.

Do not create placeholder code.

Do not leave TODO implementations.

Code Quality Rules

Required

Strong typing

Protocol-oriented design

Dependency injection

Unit-testable code

Documented public interfaces

Prohibited

Force unwraps

Temporary patches

Architecture shortcuts

Hardcoded credentials

Unused code

Deprecated APIs

Build Rule

After every implementation:

Run

xcodebuild

Analyze all errors.

Fix root causes.

Rebuild.

Repeat until build succeeds.

Never stop after the first successful compile if warnings remain.

Architecture Rule

Views

Display only.

ViewModels

State management only.

Services

Business logic only.

Repositories

Persistence and networking only.

API Layer

Data API communication only.

No layer crossing.

Camera Rules

Use AVFoundation.

Rear camera only.

Landscape only.

Stored image:

Always 1.0x.

Preview Zoom

2.5x

3.0x

4.0x

Preview only.

Never modify saved image scale.

Torch

Default 70%.

Adaptive adjustment allowed.

Flash not allowed.

Position Calibration Rules

Must occur before capture workflow.

Requirements:

Opening Center Error < 5%

Stable 3 Seconds

Automatic Advance

PTG Rules

PTG

Progressive Target Guidance

Evaluates:

- Posterior visibility
- Occlusal visibility
- Clinical usefulness

PTG does NOT primarily evaluate head rotation angle.

Levels

Dot1

Dot2

Dot3

Dot4

Dot5

Dot6

Capture Threshold

Dot4

Auto Capture Rules

Capture only when:

$PTG \geq \text{Dot4}$

AND

Stable 4 Seconds

Plateau Capture Rules

Capture when:

$PTG \geq \text{Dot4}$

AND

No visibility improvement

AND

Stable 4 Seconds

Purpose:

Reduce unnecessary retakes.

Overlay Rules

Use fixed PNG assets.

Do not dynamically generate overlays.

Do not use overlays for scoring.

Overlay purpose:

Position guidance only.

Metadata Rules

Metadata is equal in value to image data.

Every accepted image must store:

PTG

Visibility

Focus

Blur

Exposure

Calibration

Timing

Device Information

Upload Information

Dataset Integrity Rules

Every image must store:

ImageWidth

ImageHeight

ImageSizeBytes

ImageSHA256

Purpose

Duplicate detection

Integrity validation

AI export support

FileMaker Rules

Backend:

FileMaker Server 19

Use:

FileMaker Data API

Use dedicated API layouts.

Never use operational layouts for API communication.

Upload Order

Required Order

Create Session



Create Capture



Create Metadata



Upload Full Image



Upload Thumbnail



Update Session

Reason

Metadata must exist before image upload.

Offline Rules

Capture must function without network access.

Failed uploads must enter local queue.

Automatic retry required.

Retry schedule:

1 minute



5 minutes

↓

15 minutes

↓

1 hour

↓

Manual Retry

Security Rules

Required

HTTPS

TLS 1.2+

Bearer Token

Keychain Storage

Dedicated API Account

External Secure Storage

Prohibited

Guest Access

Shared Admin Accounts

Plain Text Credentials

Patient Data In Logs

Testing Rules

Every service must be testable.

Required tests:

Unit

Integration

Device

Clinical

Dataset Validation

Coverage Goal

80%+

Device Validation

Required

iPhone 11 Series

iPhone 13 Series

iPhone 15 Pro Max

Latest Supported iPhone

Definition of Done

A feature is complete only when:

Implementation Complete

Build Successful

Tests Pass

Documentation Updated

No Critical Issues

Final Principle

If a design decision improves implementation speed but reduces:

- Dataset quality
- Metadata quality
- Traceability
- Clinical usefulness

the decision must be rejected.

MonitorAI exists to create a reliable orthodontic dataset for future AI systems.

Dataset quality is the highest priority.